

Yoel Andreas Manoppo

AI Forward Deployed Engineer | Agentic AI Architecture | UI Specialist
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AI Forward Deployed Engineer on a 4-person core team building national-scale multi-agent AI for Indonesian government intelligence operations. Specializes in agentic architecture, LangGraph state machines, multi-tier search pipelines, full observability stacks, and Vue 3 tactical frontends. Track record spanning defense, blockchain, eCommerce, and digital agency environments.

WORK EXPERIENCE

AI Forward Deployed Engineer / UI Specialist at PT. GSP

May 2025 - Present

Government AI Infrastructure, Agentic Systems, Frontend Engineering

- Member of a 4-engineer team delivering production agentic AI infrastructure for BNPT (Badan Nasional Penanggulangan Terorisme) and Bareskrim at national scale.
- Built AUTOCRAWL, a 24-hour autonomous LangGraph crawler with 50 parallel workers that indexed 929 vendors overnight at 12x the original throughput estimate, using multi-tier search across DuckDuckGo, Google News RSS, and OpenSERP running Google, Bing, Yandex, and Baidu simultaneously.
- Orchestrated a 12-service Docker stack with full observability via Prometheus, Grafana, Langfuse LLM tracing, structlog, and Redis Streams rate limiting. ChromaDB vector deduplication across runs.
- Built ATLAS, a fully local voice AI pipeline for UIX intelligence using Whisper ASR with CUDA 12.x, Ollama, Groq, and edge-TTS. WebSocket live mode, persistent memory, zero cloud dependency.
- Designed Vue 3 tactical dashboards with 2.5D MapLibre GL world map, fly arcs between intelligence hubs, Apache ECharts cylinder bars per country, and a live LangGraph state machine canvas.
- Built GEOCEK OSINT geolocation engine mapping visual signals (license plates, landmarks, road features) to coordinates via Overpass API constraint filtering and weighted confidence scoring.

Software Engineer at PT. Decision Tree Indonesia

December 2025 - Present

Healthcare, eCommerce, Sales, Product

- Built CrownCare enterprise dashboard (Next.js 15, PostgreSQL, Prisma) from zero to MVP in hours with company-wide adoption in one day, reducing document turnaround time by 90 percent.
- Integrated Redis and BullMQ for async background processing (PDF generation, large video uploads) without UI blocking. Built real-time DB sync to Google Sheets for corporate reporting workflows.
- Built internal Python (Streamlit, Pandas) utilities automating TikTok Shop reporting, VLOOKUP, and SKU matching, eliminating all manual Excel work.

Lead dApp Developer (CTO) at VeChain Ecosystem

January 2024 - February 2026

Blockchain, AI, Environmental, ESG

- Architected and deployed NoPaper, ReUse, and Greencart on VeChainThor blockchain. Greencart reached 1 million active users worldwide and secured grants from VeChain Grant and Youth Climate Justice Fund.
- Built custom AI ingestion pipeline using GPT vision models to validate and extract structured data from receipts and emails with anti-fraud heuristics before minting on-chain rewards.
- Led Turborepo monorepo architecture bridging Node.js and PostgreSQL Web2 backend with Web3 on-chain data. Implemented BullMQ and Redis for high-throughput concurrent submissions.

AI Specialist at Zando Agency

May 2022 - December 2024

Digital Marketing, Supply Chain, Social Media, Agency

- Fine-tuned and deployed local LLMs via Ollama for agency-specific marketing copy ensuring full client data privacy. Automated repetitive workflows with AI scripts, eliminating thousands of manual hours per year.
- Developed and managed virtual AI influencers for fully autonomous social media campaigns using ChatGPT and generative AI tools.

KEY PROJECTS

AUTOCRAWL

LangGraph agentic 24hr crawler. 929 vendors, 50 parallel workers, 12x throughput. LangChain, Ollama, Vue 3 tactical dashboard.

GEOCEK OSINT Engine

Visual signal to coordinate geolocation. License plate mapping, Overpass API constraint filtering, GeoJSON and Folium map output.

Greencart greencart.ai

VeChain dApp with AI OCR validation and automated approval pipeline. 1 million active users worldwide.

ATLAS Voice AI

Fully local voice AI pipeline. Whisper ASR, Ollama, edge-TTS, CUDA 12.x. WebSocket live mode. Zero cloud dependency.

SENTINEL AntiFraud

Behavioral heuristic scoring. Detects synthetic vs organic activity through pattern variance analysis built from first principles.

Crowncare crowncare.site

Enterprise creator dashboard. Zero to MVP in hours, company adoption in one day, 99 percent reduction in document processing time.

TECHNICAL STACK

AI and Orchestration	LangGraph, LangChain, Ollama, OpenAI, Groq, Whisper, IBM Granite, NLLB, GPT-4o
Backend	Python, FastAPI, PostgreSQL, Redis, Docker, Playwright, Crawl4AI, Node.js, Django, SQLAlchemy
Frontend and UI	Vue 3, TypeScript, React, Next.js, TailwindCSS, MapLibre GL, Apache ECharts, Vue Flow
Observability	Prometheus, Grafana, Langfuse, ChromaDB, FlareSolvrr, OpenSERP, Structlog
Blockchain	VeChain, Solidity, Hardhat, Web3
Infrastructure	CUDA 12.x, AWS, Docker, Ubuntu, OSINT, OSM

EDUCATION AND CERTIFICATIONS

University of the People B.Sc. Business Administration	Harvard CS50 Series Computer Science Fundamentals	EFSET C2 English Certificate Proficient in English and Indonesian and Mandarin
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